COSC 325 Course Project: Motivation
Team number: 7
Team name: OpenField
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1. **Who is your target audience/end user?**

Our audience is college football fans and analysts, along with sports reporters and broadcasters. These are people who want to use information to better understand and evaluate upcoming games. The system is made for people who care about team performance and predictions, but who may not have the time or the statistical background to build a predictive model themselves.

2. **What is the problem to solve?**

We want to create a machine learning model that predicts who will win a college football game using data that is known before the game starts. The model will look at team and game data like how well a team scores, how well they stop the other team, how often they lose the ball, and how they do running and throwing the ball. It will also use what happened in previous games and maybe rankings or other things known before kickoff and use all of that to figure out which team is more likely to win.

3. **Why is this an important problem that needs to be solved right now? How will solving it improve your end user's life?**

College football generates a lot of stats every week, and it can be hard for fans to know which of those numbers actually help predict what will happen in a game. Our project aims to bring all these statistics into one prediction. Instead of just using win-loss records or average points scored, we can use a model that looks at many parts of team performance at once. This gives fans a starting point for looking at upcoming games with real numbers, makes a big pile of football data easier to understand, shows which team stats are most helpful for predictions, and lets users compare the model's guesses against what actually happened.

4. **What has been tried so far? How do these solutions fall short? Provide 2-3 full citations.** ***(For example, maybe an existing model does not look at a certain feature you think is important to consider, or maybe it is a good solution, but for a slightly different problem.)***

People have been predicting college football for a long time. Older work mostly built ranking systems and picked whichever team ranked higher, and newer work trains models on past team stats. Both show that data known before kickoff really does help, but the studies we found each leave something open.

Delen, Cogdell, and Kasap (2012) compared neural networks, decision trees, and support vector machines and got over 85% accuracy with decision trees. The catch is that they only looked at bowl games, which are very different compared to regular season games. It doesn't tell us much about a regular Saturday in October, which is what fans actually care about. South and Egros (2020) were

the closest to what our project aims to do. They trained several models on the 2011–2014 seasons and correctly picked over 70% of the winners in 2015. But they predicted the final score difference instead of just winning or losing, and their data stopped before conference realignment and the expanded playoff. We're doing the win/lose version and testing on the 2025–26 season, and we'll see if these patterns still hold today.

Delen, D., Cogdell, D., & Kasap, N. (2012). A comparative analysis of data mining methods in predicting NCAA bowl outcomes. *International Journal of Forecasting, 28*(2), 543–552. https://doi.org/10.1016/j.ijforecast.2011.05.002

South, C., & Egros, E. (2020). Forecasting college football game outcomes using modern modeling techniques. *Journal of Sports Analytics, 6*(1), 25–33. https://doi.org/10.3233/JSA-190314

5. **What machine learning technique(s) are you planning to use to solve your problem? Why?**

We plan to use supervised learning, since we will be training on labeled data where the winner of each past game is already known. We will use classification methods because we are determining a yes/no outcome: will a team win or lose?

6. **What is the dataset for your project? How much effort will it take to get the data ready for ML training?**

Our dataset is the "College Football Game Stats | 2002 to January 2026" dataset from Kaggle. It will take a good bit of effort to get it ready. Conference changes throughout the years make it hard to weigh in-conference games, and since the AP Poll only ranks 25 teams, most teams won't have an assigned ranking. We will also need to figure out how to properly weigh early season "cupcake" games. About 11% of the rows don't have a value for the 'conf_away' column, so we'll need a way to address that, and handling and filling in other missing game stats will be a challenge too.

7. **What are the baseline model(s) and paper(s) you will use for inspiration?**

We will start with a logistic regression model and use it to hone our feature selection and data preprocessing. Once we've determined the best features and preprocessing for our dataset, we'll move on to more advanced models or ensemble methods. South and Egros (2020) backs up starting simple. They found that out of the eight models they tested, a fairly basic regression model did just as well as the complicated ones.

8. **How will you measure success?** ***(Examples: Classification: Accuracy, F1, ROC-AUC, Precision/Recall; Regression: MSE, RMSE, R^2)***

Because our main task is classification, our evaluation will center on a handful of metrics. Accuracy is the percentage of games where the model correctly predicts the winner. Precision tells us how often a predicted win for a team results in a win, and recall tells us how many of the actual wins the model manages to detect. F1-score balances those two, and ROC-AUC shows how well the model separates the two outcomes. Our main metric will probably be accuracy, since the goal is simply to

predict whether a team wins or loses, but we will report several metrics instead of depending on accuracy alone because that helps us understand how the model behaves across different situations.

We will also compare every machine learning model to our baseline. Success will mean showing improvement over that baseline on a test set we did not use during training. Since the dataset only goes up to January 2026, we will train and validate on all seasons before 2025–2026 and then test the model against the entire 2025–2026 season to see how it performs and get our success measurements.

9. **Do you have enough computational resources to handle the machine learning technique and the data? Can you train the model with the resources you have?**

Yes. Kenneth has a dedicated machine at home where models can be trained, so if a model takes up a significant amount of resources, it can be trained and run there without affecting anyone. Our dataset also holds thousands of games, not millions or billions, so it's small compared to what usually gets used in machine learning. If we optimize the model and use the right techniques, it shouldn't be too computationally costly, and our main challenge will be cleaning the data and building features rather than the hardware.

10. **What are the key milestones in your roadmap?**

	Milestone/Feature Description	**Priority**	**Deadline**
1	**Explore Dataset:** Spend time looking through the dataset and document what may or may not be useful. Leverage visualization to aid in discovery.	**Medium**	**09/27/26**
2	**Data Preprocessing:** Use documentation from dataset exploration to clean the data, extract features, encode categorical features, and scale real-valued features. This begins building the ML pipeline.	**Medium**	**10/03/26**
3	**Train Initial Model:** Train a logistic regression baseline on the preprocessed data. Split off the pre-2025 seasons for training and validation and record the success metrics.	**High**	**10/07/26**
4	**Troubleshoot Initial Model:** Review where the baseline gets games wrong and figure out why. Revisit feature selection and missing value handling, then retrain to see what actually moves the metrics.	**Low**	**10/17/26**
5	**Write Midterm Project Report:** Write up the problem, the data preprocessing steps, the baseline model, and the results so far.	**High**	**10/23/26**

6	**Submit midterm project report (CP#2)**	**High**	**10/23/26**
7	**Data Preprocessing #2 and New Model Exploration:** Apply what we learned from the baseline to improve the pipeline, adding or dropping features as needed. Find out which advanced models are the best fit for our dataset.	**Low**	**10/30/26**
8	**Train New Model:** Train the models chosen in the previous step on the updated pipeline. Tune the parameters and compare each one against the logistic regression baseline.	**Medium**	**11/01/26**
9	**Troubleshoot:** Look into any model that underperforms the baseline. Adjust preprocessing, features, or parameters and retrain.	**Medium**	**11/07/26**
10	**Make Final Project Presentation:** Build the slides covering our problem, data, methods, and results. Focus on the comparison between the baseline and our final model and on which team stats mattered most.	**High**	**11/13/26**
11	**Submit project presentation (CP#3)**	**High**	**11/13/26**
12	**Report Rough Draft:** Write a complete draft of the paper in research conference format, including methods, results, and discussion. Run the final model on the 2025–2026 test season and include those numbers.	**Medium**	**11/21/26**
13	**Report Final Draft:** Revise the rough draft. Clean up figures and tables, and make sure the writing is consistent throughout.	**High**	**11/30/26**
14	**Submit final project report (CP#4)**	**High**	**12/01/26**